parameter have been moved into utility functions. The two new functions and the new version of `musicians()` are shown in the following listing.

Listing E.12 Query parameter validation utilities and updated `musicians` view

```
# RiffMates/bands/views.py
from bands.models import Band, Musician, Room, Venue
from django.core.paginator import Paginator
from django.shortcuts import get_object_or_404, render

def _get_items_per_page(request):
    # Determine how many items to show per page, disallowing <1 or >50
    items_per_page = \
        int(request.GET.get("items_per_page", 10))
    if items_per_page < 1:
        items_per_page = 10
    if items_per_page > 50:
        items_per_page = 50

    return items_per_page

def _get_page_num(request, paginator):
    # Get current page number for Pagination, using reasonable defaults
    page_num = int(request.GET.get("page", 1))

    if page_num < 1:
        page_num = 1
    elif page_num > paginator.num_pages:
        page_num = paginator.num_pages

    return page_num

def musicians(request):
    all_musicians = Musician.objects.all().order_by("last_name")
    items_per_page = _get_items_per_page(request)
    paginator = Paginator(all_musicians, items_per_page)
    page_num = _get_page_num(request, paginator)
    page = paginator.page(page_num)

    data = {
        "musicians": page.object_list,
        "page": page,
    }

    return render(request, "musicians.html", data)
```

Shared function to get the query parameter that stores the number of items to display on the page

Query parameters are strings; convert the value to an integer, and set a default value of 10.

Validate that the number of items per page is an acceptable value.

Shared function to get the query parameter that stores the page number to be displayed.

Default to page 1, and convert the parameter to an integer.

An updated version of musicians() that uses the new shared functions

Call the shared function to get the number of items per page.

Call the shared function to get the page number.

Exercise 2

Using the musician index and detail pages as a reference, create similar pages for your `Band` objects. Inside the template, use the `musicians` relationship to show the members of the band.